

MAP OPERATIONS SPECIALIST / ENGINEER

Map Data Quality - OpenStreetMap - Just-in-time Navigation Alerts

JOB OVERVIEW

Position	MAP OPERATIONS SPECIALIST
Focus	Map Data Quality, Routing & Just-in-time Alerts
Department	TASCO.COM.VN MAPPING
Location	TASCO Building Hanoi, Pham Hung
Level	Individual Contributor
Reports to	Head of Mapping

1. JOB PURPOSE

The MAP OPERATIONS SPECIALIST owns the quality of the digital map system that powers our navigation ecosystem. This role sets data-quality standards for OpenStreetMap (OSM), leads digitization of Vietnam's traffic-sign network, and tunes the routing engine to deliver accurate just-in-time alerts for millions of trips per day. This is a high-impact role that works across Engineering, Data Science, and Product to ensure the navigation experience meets international standards.

2. KEY RESPONSIBILITIES

A. Map Data Quality Governance (OpenStreetMap)

- Own and define data-quality standards for OpenStreetMap (OSM) data across all operating regions.
- Design and maintain large-scale data monitoring, cleaning, and QA pipelines.
- Resolve issues in road-network topology, node connectivity, and missing/incorrect tags to keep data routing-ready for Valhalla, GraphHopper, and OSRM.
- Define and track data-quality KPIs by region and road class.

B. Traffic Signs & Restrictions Management

- Lead digitization and continuous update of Vietnam's traffic-sign network: speed limits, enforcement cameras, turn/parking/one-way restrictions, lane info, and warning signs.
- Manage complex traffic rules: turn restrictions, time-based restrictions, and road closures (segment / route / zone).
- Ensure sign data is accurately map-matched into the routing graph to prevent routing into prohibited roads.
- Establish data-collection workflows (cameras, dashcams, GPS probes, user reports) and a continuous update pipeline.

C. Just-in-time Alerts & Navigation Experience

- Build logic and manage spatial data so the system emits accurate real-time alerts (speed limits, cameras, route changes) as vehicles move.
- Partner with Engineering to optimize map-matching (HMM, snap-to-road), minimizing latency and false positives / false negatives.
- Define SLOs/SLAs for alert accuracy and latency; set up monitoring dashboards.

D. Troubleshooting, RCA & Routing Tuning

- Lead resolution of critical positioning/navigation incidents from systems and users (GPS drift, on-ground sign mismatches).
- Conduct root-cause analysis (RCA) for trips with wrong ETA or suboptimal routes.
- Tune routing-engine costing models, penalties, and restrictions based on real-world data.
- Translate insights into actionable improvements for product and data infrastructure.

E. Technical Leadership & Mentorship

- Mentor map operators and junior engineers; review work and raise the team's quality bar.
- Build documentation, playbooks, and standard operating procedures (SOPs) for map-data work.
- Partner with Product, Engineering, and Data Science to shape the map-quality roadmap.

3. REQUIREMENTS

Must-have

- Minimum 3 years of hands-on experience as a Map Operator, GIS Data Engineer, or equivalent at a maps, navigation, ride-hailing, or logistics company.
- Bachelor's degree in Geomatics, GIS, Computer Science, or a related field.
- Deep knowledge of Vietnam's road-traffic sign system and OpenStreetMap data structures (tags, relations, restrictions).
- Proficient in QGIS, PostgreSQL/PostGIS, and spatial data analysis tooling.

- Solid understanding of routing algorithms and at least one open-source routing engine (Valhalla, GraphHopper, or OSRM).
- Proficient in Python for ETL, automation, and internal tooling.
- Experience with map editors (JOSM, iD, or RapID) and the Overpass API.
- Strong English reading comprehension for technical documentation.

Nice-to-have

- Experience with map-matching algorithms (HMM, snap-to-road) and large-scale GPS-probe data processing.
- Familiarity with osm2pgsql, Osmium, or planet-scale OSM tooling.
- Experience with streaming/real-time systems (Kafka, Flink) for location data.
- Contributions to the OpenStreetMap community or related open-source projects.
- Background in Automotive / In-car Navigation / ADAS.

Soft Skills & Mindset

- Data-driven mindset; decisions grounded in metrics.
- Strong communication with both technical and non-technical stakeholders (Product, Operations).
- Able to balance data quality, delivery speed, and operating cost.
- Takes ownership proactively; works independently and can lead small working groups when needed.

4. SUCCESS METRICS - 6 TO 12 MONTHS

- Measurable improvement in OSM data-quality KPIs in priority regions (specific targets agreed with the manager).
- Reduce false-positive / false-negative rates of just-in-time alerts to target levels.
- Reduce incidents of routing into restricted or wrong roads.
- Deliver a scalable sign-digitization pipeline ready for new-city expansion.
- Build and roll out SOPs across the team.

5. WHAT WE OFFER

- Competitive-level salary, performance bonus, and 13th-month salary.
- Premium health insurance for self and dependents; statutory insurance per Vietnamese labor law.
- Learning budget for courses, international conferences, and professional certifications.
- Flexible / hybrid working policy.
- Direct impact on a large-scale Smart Mobility / Automotive product.
- Strong engineering culture working on core map and navigation challenges.
- [Other company benefits]

6. HOW TO APPLY

Please send your email with title [Tasco Mapping Application] and CV (English or Vietnamese), together with a portfolio or descriptions of map / navigation projects you have worked on, to:

phung.dm@tasco.com.vn and tuyendung@vtii.vn

OSM contribution links or personal repositories are encouraged.